

ZKSM83: Native Lookup Proofs for SM83 Execution

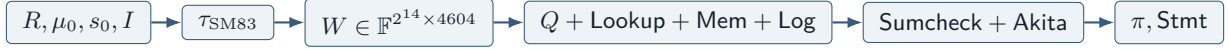
Transparent lattice-backed receipts for the DMG/MBC3 machine

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Abstract

ZKSM83 proves CPU-visible execution of an SM83 machine directly. A segment packs at most four instructions per row into $N = 2^{14}$ rows, commits $C = 4604$ multilinear columns, and checks $M = 11458$ identities of degree at most 23 by sumcheck and a transparent Akita polynomial commitment scheme. ISA, byte-operation, ROM, memory, device, log, and segment-boundary claims share one committed trace. Version 2 is transparent, not witness-hiding; no current end-to-end packed-proof measurement is claimed.



1 Proposition

Let

$$s_t = (x_t, d_t, m_t, c_t) \in \underbrace{\mathcal{X}_{\text{CPU} + \text{MBC3}}}_{21 \text{ scalars}} \times \underbrace{\mathcal{D}_{\text{DMG}}}_{17 \text{ scalars}} \times \mathcal{M}_{2^{17}} \times \mathbb{N}^4, \quad |s_t|_{\text{public}} = 38. \quad (1.1)$$

Here $c_t = (c_t^{\text{in}}, c_t^{\text{out}}, c_t^{\text{bus}}, c_t^{\text{isa}})$ and

$$(s_{t+1}, \epsilon_t) = \tau_{R,I}(s_t), \quad \epsilon_t = (\epsilon_t^{\text{bus}}, \epsilon_t^{\text{in}}, \epsilon_t^{\text{out}}, \epsilon_t^{\text{isa}}). \quad (1.2)$$

The public statement is

$$\text{Stmt} = (\nu, D_{\text{back}}, p, |R|, C_R, B_0, B_L, L, T, A, C, \mathbf{n}), \quad (1.3)$$

$$\mathbf{n} = (n_{\text{bus}}, n_{\text{in}}, n_{\text{out}}, n_{\text{isa}}), \quad (1.4)$$

$$\text{id}(\text{Stmt}) = \text{H}(\text{zksm83/native-statement/v2} \parallel \text{Enc}(\text{Stmt})), \quad (1.5)$$

where L, T, A, C are segment, source-transition, active-row, and machine-cycle counts.

Definition 1 (Implemented language).

$$\mathcal{L}_{\text{SM83}} = \{\text{Stmt} : \exists \omega \quad \mathcal{R}_{\text{row}}(\omega) \wedge \mathcal{R}_{\text{lookup}}(\omega) \wedge \mathcal{R}_{\text{mem}}(\omega) \wedge \mathcal{R}_{\text{log}}(\omega) \wedge \mathcal{R}_{\text{chain}}(\omega, \text{Stmt}) = 1\}. \quad (1.6)$$

The semantic boundary is

$$\mathcal{R}_{\text{proved}} = \{\text{SM83}, \text{MBC3-RTC}, \text{timer}, \text{PPU timing/IRQ}, \text{DMA}, \text{serial}, \text{joypad}, \text{APU/MMIO}\}, \quad (1.7)$$

$$\mathcal{R}_{\text{excluded}} = \{\text{pixels}, \text{audio samples}, \text{physical-chip provenance}, \text{ROM ownership}, \text{witness hiding}\}. \quad (1.8)$$

The SM83 timing and opcode model follows the public hardware references in Pan Docs [3, 4]; the proof architecture is lookup-centric in the sense of Jolt and Lasso [1, 9].

2 Packed trace

One segment uses the Boolean hypercube

$$\mathcal{H} = \{0, 1\}^{14}, \quad N = |\mathcal{H}| = 16384. \quad (2.1)$$

For row $r \in \mathcal{H}$, let a_r be active, i_r be an instruction-block tag, $\ell_{r,j}$ be lane activity, and $e_{r,k}$ be a machine-event selector:

$$a_r(a_r - 1) = i_r(i_r - 1) = \ell_{r,j}(\ell_{r,j} - 1) = e_{r,k}(e_{r,k} - 1) = 0, \quad (2.2)$$

$$\ell_{r,0} = i_r, \quad \ell_{r,j+1}(1 - \ell_{r,j}) = 0, \quad (2.3)$$

$$a_r = i_r + \sum_k e_{r,k}, \quad i_r \sum_k e_{r,k} = 0, \quad (2.4)$$

$$1 \leq \sum_{j=0}^3 \ell_{r,j} \leq 4 \quad (i_r = 1). \quad (2.5)$$

Resource ownership is explicit. For $S_M = 6$ machine-cycle slots, $S_B = 5$ bus slots, and $\omega_{r,s,j}^X \in \{0, 1\}$:

$$\sum_{j=0}^3 \omega_{r,s,j}^X \in \{0, 1\}, \quad X \in \{M, B\}, \quad 0 \leq s < S_X, \quad (2.6)$$

$$(1 - \ell_{r,j}) \sum_s \omega_{r,s,j}^X = 0, \quad (2.7)$$

$$\ell_{r,j} \prod_{q=1}^{S_X} \left(\sum_s \omega_{r,s,j}^X - q \right) = 0. \quad (2.8)$$

The five local boundaries $x_{r,0}, \dots, x_{r,4}$ are shared across four lanes; the device boundary (d_r^-, d_r^+) is shared across the row:

$$\ell_{r,j}(x_{r,j+1} - \tau_{or,j}^{\text{cpu}}(x_{r,j}, b_{r,j})) = 0, \quad (2.9)$$

$$(1 - \ell_{r,j})(x_{r,j+1} - x_{r,j}) = 0, \quad (2.10)$$

$$e_{r,k}(d_r^+ - \Delta_k(d_r^-, b_r, I_r)) = 0. \quad (2.11)$$

For lane j , local bus slot u , shared slot $s \geq u$, define $k = s - u$ and

$$f_{r,j,k} = \omega_{r,k,j}^B \begin{cases} 1, & k = 0, \\ 1 - \omega_{r,k-1,j}^B, & k > 0, \end{cases} \quad (2.12)$$

$$m_{r,j,u,s} = \begin{cases} f_{r,j,k}, & u = 0, \\ f_{r,j,k} \omega_{r,s,j}^B, & u > 0, \end{cases} \quad (2.13)$$

$$b_{r,j,u} = \sum_{s=u}^4 m_{r,j,u,s} b_{r,s}. \quad (2.14)$$

The $4 \sum_{u=0}^4 (5 - u) = 60$ values $m_{r,j,u,s}$ are committed columns. This replaces repeated cubic routing expressions by degree-one consumers.

The complete column geometry is

$$C = 3542 + 806 + 256 = 4604, \quad (2.15)$$

$$806 = \underbrace{5 \cdot 106}_{\text{shared boundaries}} + \underbrace{4 \cdot 54}_{\text{lane auxiliaries}} + \underbrace{60}_{\text{bus matches}}, \quad (2.16)$$

$$256 = \underbrace{4}_{\text{lanes}} \cdot \underbrace{2}_{\text{execution queries}} \cdot \underbrace{(1 + 19 + 12)}_{\text{selector+address+output}}. \quad (2.17)$$

3 Row relation

Let $W_r \in \mathbb{F}^{4604}$. The deepest relation is

$$Q(W_r) = (Q_0(W_r), \dots, Q_{11457}(W_r)) = \mathbf{0}, \quad (3.1)$$

Relation prefix	columns	identities	degree
Routing	50	129	7
Boundary	145	297	3
Front end	729	1096	7
Machine	796	1353	7
Mutable-memory row relation	3542	8257	15
Packed CPU plus execution lookup	4604	11458	23

Table 1: Source-asserted prefix geometry. Prefix rows share columns; counts are not additive.

$$Q = Q_{\text{route}} \cup Q_{\text{boundary}} \cup Q_{\text{ISA}} \cup Q_{\text{bus}} \cup Q_{\text{flow}} \cup Q_{\text{machine}} \cup Q_{\text{device}} \cup Q_{\text{MMIO}} \cup Q_{\text{mem}} \cup Q_{\text{CPU}} \cup Q_{\text{exec-wf}}, \quad (3.2)$$

$$\max_j \deg Q_j \leq 23. \quad (3.3)$$

Byte and word reconstruction uses Boolean limbs:

$$b_k(b_k - 1) = 0, \quad v = \sum_{k=0}^7 2^k b_k, \quad (3.4)$$

$$w_k(w_k - 1) = 0, \quad u = \sum_{k=0}^{15} 2^k w_k. \quad (3.5)$$

Typical selector-gated identities are

$$q_{\text{add}}[r - a - b + 2^8 c] = 0, \quad (3.6)$$

$$q_{\text{sub}}[a - b - r - 2^8 c] = 0, \quad (3.7)$$

$$q_{\text{jump}}[pc^+ - (q_t t + (1 - q_t) pc_{\text{seq}})] = 0, \quad (3.8)$$

$$q_{\text{stack}}[sp^+ - (sp + \delta_{\text{sp}})] = 0. \quad (3.9)$$

Flags, DAA, CB-prefixed operations, interrupts, HALT, MBC3 bank selection, timer stages, PPU/STAT ordering, DMA, serial, joypad, APU, and typed MMIO instantiate the same pattern $q \cdot f(W_r) = 0$.

The fixed tables are

$$T_{\text{ISA}} : [2^9] \rightarrow \mathbb{F}^{38}, \quad T_{\text{exec}} : [2^{19}] \rightarrow \mathbb{F}^{12}, \quad R : [2^{20}] \rightarrow \mathbb{F}. \quad (3.10)$$

There are four ISA queries, at most eight byte-operation queries, and at most five ROM bus queries per packed row.

4 Shout-style lookup equations

For a table $T : \{0, 1\}^m \rightarrow \mathbb{F}^v$, selected trace rows $x \in \mathcal{H}$, address bits $A(x) \in \{0, 1\}^m$, selector $q(x)$, and output $y(x)$, sample γ, λ and compress

$$T_\gamma(u) = \sum_{h=0}^{v-1} \gamma^h T_h(u), \quad y_\gamma(x) = \sum_{h=0}^{v-1} \gamma^h y_h(x), \quad (4.1)$$

$$\chi_{A(x)}(u) = \prod_{j=0}^{m-1} (A_j(x) u_j + (1 - A_j(x))(1 - u_j)), \quad (4.2)$$

$$R_A(u) = \sum_{x \in \mathcal{H}} \lambda_x q(x) \chi_{A(x)}(u). \quad (4.3)$$

Then

$$\boxed{\sum_{u \in \{0, 1\}^m} R_A(u) T_\gamma(u) = \sum_{x \in \mathcal{H}} \lambda_x q(x) y_\gamma(x)}. \quad (4.4)$$

The left side is a table product sumcheck; the construction of R_A is an address-binding sumcheck. Equation (4.4) is instantiated for

$$\text{ISA : } (m, v) = (9, 38), \quad 4 \text{ lanes}, \quad (4.5)$$

$$\text{Exec : } (m, v) = (19, 12), \quad 8 \text{ slots}, \quad (4.6)$$

$$\text{ROM : } (m, v) = (20, 1), \quad 5 \text{ bus slots}. \quad (4.7)$$

All addresses, outputs, selectors, table commitments, and openings are Fiat-Shamir bound.

5 Mutable memory

For a selected event $e = (a, v^-, v^+, t^-, t)$ with write selector w_e , choose $\alpha, \beta \leftarrow \mathbb{F}$ and define

$$u_e = a + \alpha(w_e v^- + (q_e - w_e) v^+) + \alpha^2 t^-, \quad (5.1)$$

$$v_e = a + \alpha v^+ + \alpha^2 t, \quad (5.2)$$

$$(\beta - u_e) z_e^{\text{in}} = q_e, \quad (\beta - v_e) z_e^{\text{out}} = q_e. \quad (5.3)$$

With $\langle f \rangle_n = 2^{-n} \sum_{x \in \{0,1\}^n} f(x)$ and boundary inverse-table averages Z_0, Z_L :

$$\boxed{2^{17}(Z_0 - Z_L) + 2^{14} \sum_{s=0}^4 (\langle z_s^{\text{out}} \rangle_{14} - \langle z_s^{\text{in}} \rangle_{14}) = 0.} \quad (5.4)$$

Fixed clocks, predecessor timestamps, initial values, final values, and Equation (5.4) bind the 2^{17} -byte mutable image.

6 Ordered logs and continuity

For $k \in \{\text{bus}, \text{in}, \text{out}, \text{isa}\}$, compress tuple $v = (v_0, \dots, v_{d_k-1})$ as

$$\phi_k(v) = \sum_{j=0}^{d_k-1} \gamma_{k,j} v_j, \quad z_{k,e} = \frac{q_{k,e}}{\zeta_k - \phi_k(v_e)}. \quad (6.1)$$

Canonical positions are tuple fields. For trace inverse columns Z_k^{tr} and table inverse columns Z_k^{tab} :

$$2^{14} \sum_e \langle Z_{k,e}^{\text{tr}} \rangle_{14} = 2^{17} \langle Z_k^{\text{tab}} \rangle_{17}, \quad (6.2)$$

$$2^{17} \langle q_k^{\text{tab}} \rangle_{17} = n_k. \quad (6.3)$$

For state-mixing challenge ρ and inverse point ζ , define the row-tagged token

$$\Phi_\rho(s, t) = t + \sum_{j=0}^{37} \rho^{j+1} s_j. \quad (6.4)$$

With

$$z_r^+ = \frac{a_r}{\zeta - \Phi_\rho(s_r^+, r+1)}, \quad z_r^- = \frac{a_r}{\zeta - \Phi_\rho(s_r^-, r)}, \quad (6.5)$$

$$c_r = a_r - i_r + \sum_{j=0}^3 \ell_{r,j}, \quad (6.6)$$

continuity reduces to

$$\boxed{2^{14}(\langle z^+ \rangle_{14} - \langle z^- \rangle_{14}) + \frac{1}{\zeta - \Phi_\rho(s_0, 0)} - \frac{1}{\zeta - \Phi_\rho(s_L, A)} = 0,} \quad (6.7)$$

$$\boxed{2^{14} \langle c \rangle_{14} = T}, \quad A \leq T \leq 4A. \quad (6.8)$$

7 Uniform sumcheck and Akita PCS

The row identities are reduced by the sumcheck protocol [8]. For column $w_j : \{0, 1\}^{14} \rightarrow \mathbb{F}$,

$$\tilde{w}_j(z) = \sum_{x \in \{0,1\}^{14}} w_j(x) \text{eq}(z, x), \quad \text{eq}(z, x) = \prod_{k=0}^{13} (z_k x_k + (1 - z_k)(1 - x_k)). \quad (7.1)$$

Sample $r \leftarrow \mathbb{F}^{14}$ and $\eta \leftarrow \mathbb{F}^*$, and mix the identities:

$$Q_\eta(W(x)) = \sum_{j=0}^{M-1} \eta^j Q_j(W(x)), \quad (7.2)$$

$$0 = \sum_{x \in \{0,1\}^{14}} \text{eq}(r, x) Q_\eta(W(x)). \quad (7.3)$$

Sumcheck yields $u \in \mathbb{F}^{14}$ and

$$\text{claim}_{14} = \text{eq}(r, u) Q_\eta(\tilde{W}_0(u), \dots, \tilde{W}_{C-1}(u)). \quad (7.4)$$

Since $\deg Q \leq 23$, equality weighting gives round degree at most 24 and

$$n_{\text{eval/round}} = 25, \quad n_{\text{rounds}} = 14, \quad n_{\text{full row eval}} = 25(2^{14} - 1) = 409575. \quad (7.5)$$

Pad the witness with four zero columns:

$$\bar{W} = (W_0, \dots, W_{4603}, 0, 0, 0, 0) \in \mathbb{F}^{2^{14} \times 4608}, \quad (7.6)$$

$$G_g = (\tilde{W}_{128g}, \dots, \tilde{W}_{128g+127}), \quad 0 \leq g < 36, \quad (7.7)$$

$$C_g = \text{Com}_{\text{Akita}}(G_g), \quad (7.8)$$

$$\pi_i^{\text{open}} = \text{Open}_{\text{Akita}}(G_{2i}, G_{2i+1}; u), \quad 0 \leq i < 18. \quad (7.9)$$

Akita is a transparent lattice PCS whose binding target is stated under Module-SIS [5, 6]:

$$A \xleftarrow{\$} R_q^{n \times m}, \quad \text{MSIS}_{n,m,q,\beta} : \quad 0 \neq z \in R^m, \quad Az = 0 \pmod{q}, \quad \|z\| \leq \beta. \quad (7.10)$$

Nightstream [7] is relevant as an implementation reference for lattice-based CCS folding architecture, a separated paper-exact versus optimized oracle/sumcheck path, and seeded Ajtai commitment engineering. It is not used here as a production backend and is not evidence of end-to-end SM83 or Game Boy proving throughput.

The transcript binds

$$\text{tr} = \text{H}(\nu \parallel D_{\text{back}} \parallel D_{\text{schedule}} \parallel C_0 \parallel \dots \parallel C_{35} \parallel \text{Stmt} \parallel \text{domain}), \quad (7.11)$$

using Blake2b-512 internally for Fiat-Shamir. The wire cut is strict:

$$\text{Decode}(\text{ZKSM83R2}) = \nu_2, \quad \text{Decode}(\text{ZKSM83R1}) = \perp. \quad (7.12)$$

Fiat-Shamir compilation follows the standard transform paradigm [2].

8 Receipt composition

One segment proof is

$$\pi_k = (C_W, \pi_Q, \pi_{\text{ISA}}, \pi_{\text{exec}}, \pi_{\text{ROM}}, \pi_{\text{mem}}, \pi_{\text{cont}}, \pi_{\text{log}}). \quad (8.1)$$

Before any folding layer is introduced, the implementation types each obligation by family and evaluation binding: CPU and device row relations use the packed-row sumcheck terminal point; ISA and execution lookups use shared witness openings; ROM uses a table opening; mutable memory uses the shared witness plane; continuity uses segment-boundary challenges; and logs use ordered-log challenges. Accumulation may only combine claims with the same family and evaluation binding.

For $0 \leq k < L$:

$$B_k^+ = B_{k+1}^-, \quad C_{\mu,k}^+ = C_{\mu,k+1}^-, \quad (8.2)$$

$$\sum_k T_k = T, \quad \sum_k A_k = A, \quad \sum_k C_k = C, \quad \sum_k \mathbf{n}_k = \mathbf{n}. \quad (8.3)$$

The verifier accepts exactly when

$$\text{Verify}(\text{Stmt}, \pi) = \bigwedge_{k=0}^{L-1} [\text{Verify}_Q \wedge \text{Verify}_{\text{ISA}} \wedge \text{Verify}_{\text{exec}} \wedge \text{Verify}_{\text{ROM}} \wedge \text{Verify}_{\text{mem}} \wedge \text{Verify}_{\text{cont}} \wedge \text{Verify}_{\text{log}}]_k \wedge \mathcal{R}_{\text{chain}}. \quad (8.4)$$

No emulator, ROM image, memory image, or trace is a verifier input.

9 Conditional soundness

Proposition 1 (Composition). *Assume Akita evaluation binding, sumcheck soundness, lookup and log-derivative soundness, collision resistance of the canonical hashes, and a random-oracle Fiat-Shamir transform. Then*

$$\Pr [\text{Verify}(\text{Stmt}, \pi) = 1 \wedge \text{Stmt} \notin \mathcal{L}_{\text{SM83}}] \leq \varepsilon_{\text{PCS}} + \varepsilon_{\text{SC}} + \varepsilon_{\text{lookup}} + \varepsilon_{\text{log}} + \varepsilon_{\text{hash}} = \text{negl}(\lambda). \quad (9.1)$$

Reduction skeleton.

$$\begin{aligned} \text{Verify}_{\text{PCS}} = 1 &\implies \exists W : \widetilde{W}(u) = v_u, \\ \text{Verify}_Q = 1 &\implies \forall r < N, Q(W_r) = 0, \\ \text{Verify}_{\text{lookup}} \wedge \text{Verify}_{\text{mem}} \wedge \text{Verify}_{\text{log}} = 1 &\implies (R, \mu_0, \mu_L, \epsilon_{0:T}) \text{ are commitment-bound,} \\ \text{Verify}_{\text{cont}} \wedge \mathcal{R}_{\text{chain}} = 1 &\implies s_0 \xrightarrow{\tau} \dots \xrightarrow{\tau} s_T, \quad (s_0, s_T, T, A, C, \mathbf{n}) = \text{Stmt}. \end{aligned}$$

Any false implication breaks one assumed component. $\square$

10 Static cost and evidence boundary

The logical witness sizes are exact source-level quantities:

$$|W|_{u64} = 4604 \cdot 2^{14} \cdot 8 = 603455488 \text{ bytes} = 575.5 \text{ MiB}, \quad (10.1)$$

$$|W|_{\mathbb{F}_{128}} \approx 4604 \cdot 2^{14} \cdot 16 = 1.124 \text{ GiB}, \quad (10.2)$$

$$N_{\text{slot}} = 409575 \cdot 11458 = 4692910350. \quad (10.3)$$

N_{slot} counts constraint-slot evaluations in the main sumcheck model, not native CPU instructions or measured field operations.

A benchmark report must separate at least the following buckets before making a performance claim:

$$\begin{aligned} &\text{WitnessBuild, RelEval, Commit, ROM, Mem, Cont, Log,} \\ &\text{Sumcheck, Open, Encode, Verify, } |\pi|. \end{aligned} \quad (10.4)$$

The current implementation exposes process-local diagnostics for the proof components in Equation (10.4), except that witness construction and proof-byte attribution still require a dedicated benchmark harness.

For T source transitions and average packing density $1 \leq \bar{\rho} \leq 4$,

$$A \approx \left\lceil \frac{T}{\bar{\rho}} \right\rceil, \quad L \approx \left\lceil \frac{T}{2^{14} \bar{\rho}} \right\rceil, \quad \text{Cost}_P = L (\text{Commit}_{36} + \text{Open}_{18} + \text{SC}_Q + \text{Lookup} + \text{Mem} + \text{Log}). \quad (10.5)$$

At snapshot 1062c35:

$$\underbrace{(\mathcal{R}_{\text{row}}, \mathcal{R}_{\text{lookup}}, \mathcal{R}_{\text{mem}}, \mathcal{R}_{\text{log}}, \mathcal{R}_{\text{chain}}, \text{Prover}, \text{Verifier})}_{\text{connected in source}} = \checkmark, \quad (10.6)$$

$$\underbrace{(\text{packed e2e, cartridge chain, benchmark})}_{\text{current evidence}} = ?, \quad (10.7)$$

$$\text{WitnessHiding} = 0. \quad (10.8)$$

Thus Equations (10.3)–(10.5) are models, not measurements.

11 Code correspondence

Object	Implementation
$\tau_{R,I}$	<code>zksm83-core::StepRelation</code>
W, Q	<code>block_cpu::BlockCpuWitness, BlockCpuRelation</code>
$T_{\text{ISA}}, T_{\text{exec}}, R$	<code>isa_lookup, execution_lookup, rom_lookup</code>
$\mathcal{R}_{\text{mem}}, \mathcal{R}_{\text{log}}$	<code>memory, logs</code>
$\mathcal{R}_{\text{cont}}, \mathcal{R}_{\text{chain}}$	<code>continuity, receipt</code>
$\text{SC}, \text{Com}, \text{Open}$	<code>uniform::sumcheck, uniform::commitment, pcs</code>
$\text{Bench}, \text{Obligation}$	<code>optimization, metrics, BlockCpuRelation::evaluation_profile</code>

Table 2: Equation-to-source map in `crates/zksm83-jolt/src`.

12 Conclusion

$$(R, \mu_0, s_0, I) \xrightarrow{\tau_{\text{SM83}}} W \xrightarrow{Q+\text{Shout}+\text{Mem}+\text{Log}} \pi \xrightarrow{\text{Akita}} (\text{Stmt}, \text{accept}) \quad (12.1)$$

The open empirical obligation is a current independently verified packed v2 receipt; the open privacy obligation is an explicit witness-hiding layer.

References

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